

Lionael Surya Dwitama

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PROFESSIONAL SUMMARY

Data Science & Analytics student who builds and evaluates AI-generated output hands-on — from LLM-generated explanations in a live market-analysis tool to SHAP-based model interpretability for a flood-risk predictor. Experienced in auditing AI output against accuracy and reliability standards, debugging and validating data against expected results, and writing clear, precise documentation of findings. Comfortable working independently against detailed technical specifications and maintaining consistency across repeated, structured tasks.

EDUCATION

Seneca Polytechnic

Honours Bachelor of Data Science and Analytics | Newnham Campus

2024 – Present

Toronto, ON

- Cumulative GPA: 3.9/4.0; President's Honour List (Winter 2025 and Winter 2026)
- Relevant coursework: Python Programming, SQL, Data Visualization, Statistics, Machine Learning, Database Systems

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

AI & LLM: Anthropic Claude API integration, LLM-generated content evaluation, prompt-driven explanation generation, XGBoost, SHAP explainability, Optuna

Data & Analytics: pandas, Power BI, Microsoft Excel, statistical analysis, data visualization

Tools & Platforms: Git, GitHub, GitHub Actions, VS Code, Google Workspace, Microsoft Office

PROJECTS

IDX Stock Scanner Agent — AI-Powered Market Analysis Tool

Ongoing

Personal project | github.com/suryalionael/idx-stock-scanner-agent

- Built a three-stage pipeline — rule-based filtering, XGBoost ranking, and LLM-generated explanations via the Anthropic Claude API — that turns raw market data into ranked, explained output across three trading modes.
- Ran a full technical audit of the tool's output and caught a data-authenticity issue where broker-activity figures were being silently generated from mock data instead of the live Index Alpha API; rebuilt the module against the real API and added staleness disclosures so users can trust what they see.
- Continuously evaluate model and LLM output for accuracy and reliability, distinguishing genuinely usable signal from misleading or unverified content before it reaches users.

FloodCast Jakarta — Flood Risk Prediction with Explainable ML

2026

Dicoding AI Impact Challenge 2026 | github.com/suryalionael/Datathon-Dicoding2026-FloodRisk

- Built a multi-horizon XGBoost ensemble to predict flood risk across Jakarta, tuned with Optuna and paired with SHAP explainability so every prediction traces back to the specific factors driving it.
- Developed an interactive Leaflet.js dashboard to present predictions and their underlying explanations clearly to non-technical stakeholders.

EXPERIENCE

Career Peer, Seneca Works

Sep 2026 – Present

Seneca Polytechnic, Toronto, ON

- Review student resumes and cover letters against structured evaluation criteria, providing detailed written feedback to improve clarity, accuracy, and consistency with professional standards.
- Support students with job-board access and career resources, and help run career events and workshops.

Data Entry and Web Development

Jul 2025 – Present

QuickRn Canada

- Investigate and resolve website and course-access issues through methodical troubleshooting, documenting each issue and its resolution for consistency across cases.
- Respond to customer inquiries via email with clear, professional written communication.

Software Engineering Intern

Jun 2025 – Aug 2025

CV. Veda Sakti Dharma, Jakarta, Indonesia

- Debugged software issues and validated payroll tax calculation output against expected results to improve system reliability and reporting accuracy.